

Sequences and series - Set 2

TMUA practice questions by topic. Specification MM2.1, MM2.2, MM2.3, M4.19. 11 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The sum of the first n terms of an arithmetic series is S_n . Given that $S_{10} = 145$ and $S_{20} = 490$, find the common difference.
- A 2
B 1
C 5
D 4
E 3
F $\frac{3}{2}$
- 2 A sequence is defined by $u_1 = 1$ and $u_{n+1} = u_n + 2n + 1$. Which of the following is an expression for u_n ?
- A $n^2 - n + 1$
B $2n - 1$
C $2^n - 1$
D $n^2 + 1$
E $n(n + 1)/2$
F n^2
- 3 A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?
- A P is necessary and sufficient for Q .
B P is necessary but not sufficient for Q .
C P is sufficient but not necessary for Q .
D P is neither necessary nor sufficient for Q .

- 4 An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?
- I If $d > 0$ then every term after some point is positive.
 - II The sum of the first n terms is a quadratic function of n .
 - III If a and d are integers then S_n is an integer for every n .
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
- 5 A geometric series has first term a and common ratio r with $|r| < 1$. Its sum to infinity is 8 and the sum to infinity of the squares of its terms is $\frac{64}{3}$. Find r .
- A $-\frac{1}{2}$
- B $\frac{1}{2}$
- C $-\frac{4}{3}$
- D $\frac{4}{3}$
- E $-\frac{3}{4}$
- F $\frac{3}{4}$
- 6 A sequence satisfies $x_{n+1} = \frac{1}{1-x_n}$ with $x_1 = 2$. Find x_{100} .
- A 1
- B 2
- C $\frac{1}{2}$
- D $-\frac{1}{2}$
- E 0
- F -1

- 7 The sum of the first n natural numbers is S . The sum of the first $2n$ natural numbers is kS . Which of the following is true?
- A $k = \frac{4n+2}{n}$
- B $k = 4$ for all n
- C $k = \frac{2(2n+1)}{n+1}$
- D $k = 2$ for all n
- E $k = 4 - \frac{2}{n}$
- 8 A ball is dropped from a height of 10 m. After each bounce it rises to $\frac{3}{5}$ of its previous height. Find the total distance travelled by the ball before it comes to rest.
- A 30 m
- B 20 m
- C 40 m
- D 25 m
- E 50 m
- F 35 m
- 9 The first three terms of a geometric sequence are x , $x+6$ and $x+15$. Find the fourth term.
- A $\frac{81}{25}$
- B $\frac{25}{73}$
- C $\frac{225}{73}$
- D $\frac{225}{125}$
- E $\frac{125}{4}$
- F 45
- 10 The integers from 1 to 100 inclusive are written down. What is the sum of those that are multiples of 3 or 5 (or both)?
- A 2500
- B 2350
- C 2733
- D 2103
- E 2418
- F 2318

- 11** A sequence has $u_1 = 3$ and $u_{n+1} = \frac{u_n + 5}{2}$. Which of the following statements is/are true?
- I** The sequence is increasing.
 - II** $u_n < 5$ for all n .
 - III** $u_n = 5 - 2^{2-n}$ for all n .
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III